ACADEMIC YEAR 2026

A RISK-CALIBRATED ADAPTIVE HANDOVER METHOD FOR TEACHER-GUIDED FLOW-MATCHING TEXT-TO-SPEECH INFERENCE

HANDOVER

T = 0 · NOISE

TEACHER → STUDENT

T = 1 · SPEECH

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PRESENTATION STRUCTURE

OVERVIEW OF PROPOSAL CHAPTERS

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3.7 Calibration with a guarantee

3.1–3.8 Design, workflow and benchmarks

★ Contribution and summary

App. Preliminary tests and supplementary data

SECTION 1.1 · BACKGROUND OF THE STUDY

FLOW-MATCHING TTS: COST & DISTILLATION TRADE-OFFS

FLOW MATCHING

$$dx/dt = v_\vartheta(x, t)$$

Noise moves to a mel-spectrogram step by step; **each step is one network evaluation**. F5-TTS is our teacher.

CLASSIFIER-FREE GUIDANCE (CFG)

$$v = v_u + w(v_c - v_u)$$

Conditional + unconditional prediction: **2 forward passes** per guided teacher step.

FEW-STEP STUDENT DISTILLATION

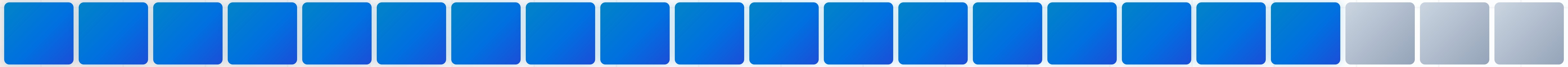
4 steps

DMOSpeech distilled the teacher into a **few-step student**, 128 → 4.

MODE SHRINKAGE & HYBRID RECOVERY

The student's prosody is less varied (**mode shrinkage**). DMOSpeech 2 lets the teacher run the first steps and the student finish.

Released sampler, switching time 0.25 🟦 teacher 18 ▨ student 3

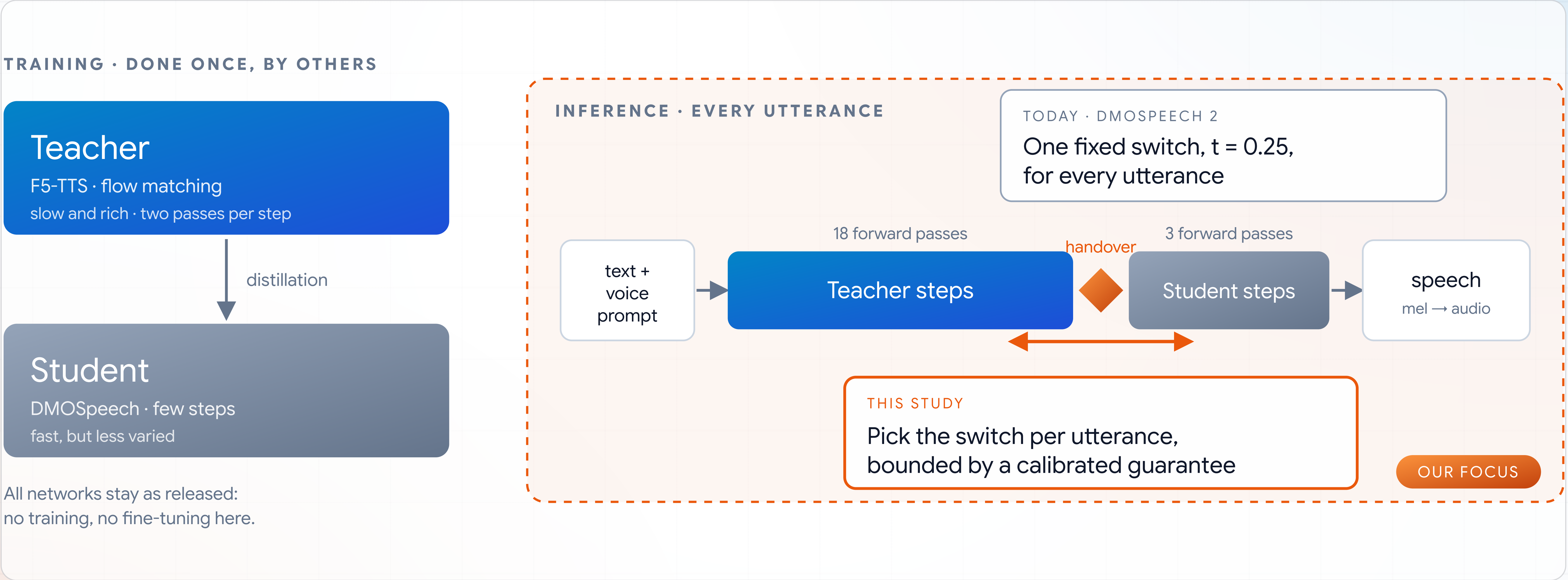


21 forward passes for **every** utterance

SOURCES Lipman et al. (2023) · Y. Chen et al. (2025), *F5-TTS* · Ho & Salimans (2022), *Classifier-free diffusion guidance* · Li et al. (2025), *DMOSpeech*, ICML · Li et al. (2026), *DMOSpeech 2*, AAAI

SECTION 1.1 · THE MODEL AT A GLANCE

TEACHER-GUIDED TTS, AND WHERE WE WORK



TRAINING · DONE ONCE, BY OTHERS

Teacher
F5-TTS · flow matching
slow and rich · two passes per step

distillation

Student
DMOSpeech · few steps
fast, but less varied

All networks stay as released:
no training, no fine-tuning here.

INFERENCE · EVERY UTTERANCE

TODAY · DMOSPEECH 2
One fixed switch, t = 0.25,
for every utterance

text + voice prompt
18 forward passes
Teacher steps
handover
3 forward passes
Student steps
speech
mel ⟶ audio

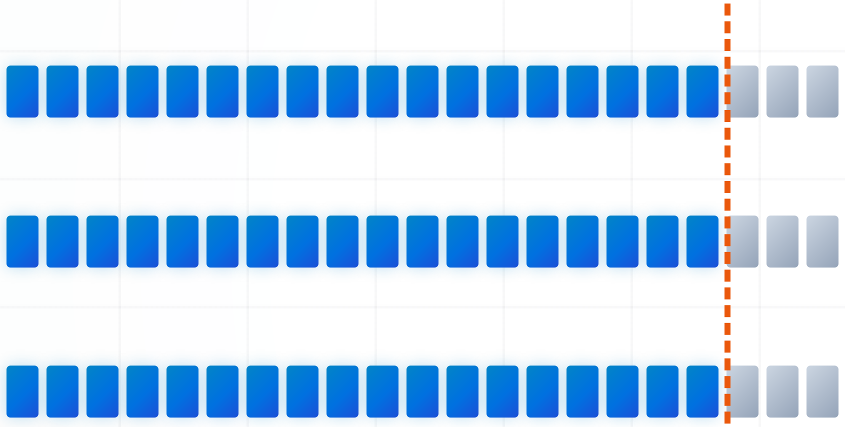
THIS STUDY
Pick the switch per utterance,
bounded by a calibrated guarantee

OUR FOCUS

SOURCES Y. Chen et al. (2025), *F5-TTS* · Li et al. (2025), *DMOSpeech* · Li et al. (2026), *DMOSpeech 2* · Proposal Sections 1.1 and 3.2 (simplified outline)

SECTION 1.2 · PROBLEM STATEMENT

LIMITATIONS OF STATIC HANDOVER SCHEDULES ACROSS INPUTS



INPUT A
clean, easy

INPUT B
moderate

INPUT C
noisy prompt

FIXED HANDOVER t_h = 0.293 · 21 PASSES EACH

Too late wastes guided passes. Too early may degrade speech. **Does every utterance need 18 teacher passes?**

PRELIMINARY T4 EVIDENCE · APPENDIX D.1 · 1,000 INSTANCES

Fixed early switches fail on **25–33%** of inputs (noisy prompts 49–64%, clean 1.0–2.4%). A per-input choice in hindsight needs **14.85 of 21** passes (−29%). No single early switch is safe for every input.

1.2.1 **Uncharacterized effects.** How the switching time changes WER, SIM, UTMOS and cost per utterance is unmeasured.

1.2.2 **The handover boundary.** Can inference-time signals pick an earlier switch without losing quality?

1.2.3 **No reliability guarantee.** Adaptive selection alone says nothing about the rate of added failures.

1.2.4 **Application constraints.** Faster synthesis is not evidence of benefit to autistic users.

SOURCES Li et al. (2026), *DMOSpeech 2*, AAAI · Ho & Salimans (2022) · Angelopoulos et al. (2025), *Learn then test*, Ann. Appl. Stat. · Proposal Appendix D.1 (development evidence)

SECTION 1.3 · OBJECTIVES OF THE STUDY

GENERAL & SPECIFIC RESEARCH OBJECTIVES

1

CHARACTERIZE

Measure how WER, SIM, UTMOS, forward passes and runtime vary across candidate switching times and inputs.

→ QUALITY–COST MAP

2

FORMULATE

Pose selection as constrained cost minimization; design a training-free policy with no extra network evaluations; prove its properties.

→ POLICY + 4 PROPOSITIONS

3

CALIBRATE

Use Learn-Then-Test to keep added-failure risk within a tolerance α at confidence $1 - \delta$; check it on held-out speakers.

→ CERTIFIED THRESHOLD $\hat{\Lambda}$

4

COMPARE

Benchmark SNR-only, stabilization-only and combined variants against fixed switches and an oracle, clean and noisy.

→ DOES EACH SIGNAL ADD VALUE?

FROM Proposal, Section 1.3 · all pretrained weights frozen; training-free

SECTIONS 1.4–1.5 · SCOPE & SIGNIFICANCE

SCOPE, DELIMITATIONS, & TARGET BENEFICIARIES

IN	**Only the switching time varies.** Restart level fixed at the released default (j = 1) as a controlled setting.
IN	Teacher, student, duration predictor and vocoder **stay frozen**.
IN	English zero-shot prompts from **Seed-TTS test-en**, clean and 15 dB noisy.
IN	All runs on an **NVIDIA Tesla T4** through Google Colab.
OUT	A per-utterance guarantee; the bound holds **on average** over inputs.
OUT	Building or testing an assistive app; clinical or learner outcomes.

WHO BENEFITS

SPEECH RESEARCHERS

A formulation, proofs and calibration for teacher-guided samplers in general.

TTS DEVELOPERS

Lower cost with a stated bound, no retraining.

ASSISTIVE-TECH DEVELOPERS

Cheaper, predictable synthesis for personalized voices.

FUTURE RESEARCHERS

A reusable protocol: disjoint calibration, fixed failure rule, exact cost.

SOURCES Anastassiou et al. (2024), *Seed-TTS* · Ardila et al. (2020), *Common Voice*, LREC · Muharib et al. (2024), JADD · Therrien et al. (2025), AAC

SECTION 2.4 · RELATED LITERATURE

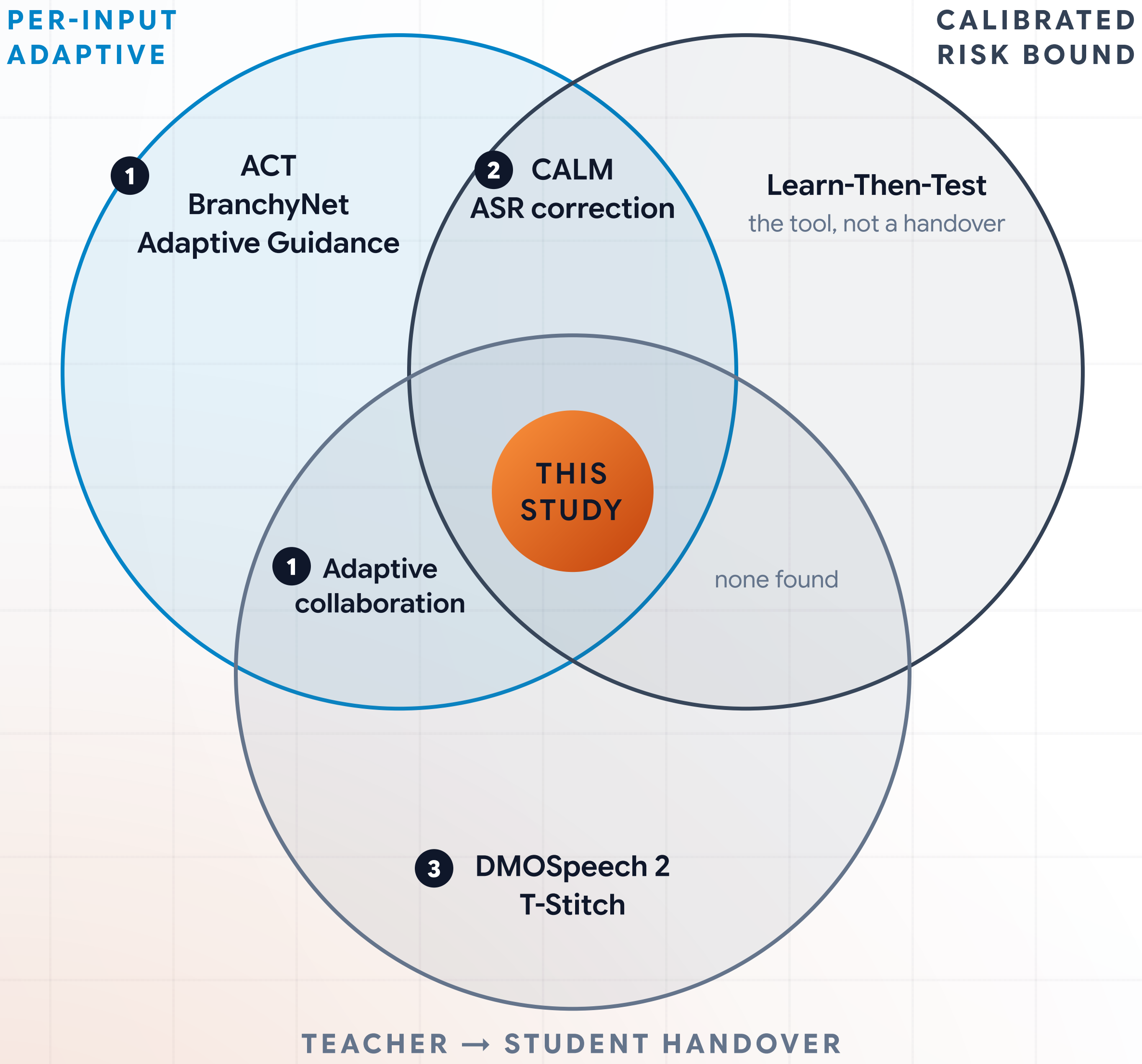
INPUT-ADAPTIVE COMPUTATION & DYNAMIC INFERENCE

METHOD	WHAT IS ADAPTED	DECISION SIGNAL	RULE	GUARANTEE
ACT Graves, 2016	Internal steps per input	Learned halting unit	Learned	NONE
BranchyNet Teerapittayanon et al., 2016	Exit layer	Classifier confidence	Tuned threshold	NONE
CALM Schuster et al., 2022	Exit layer per token	Confidence or saturation	Calibrated threshold	RISK CONTROL
Adaptive collaboration Starodubcev et al., 2024	Whether the teacher refines	Quality estimator on a full student sample	Tuned threshold	NONE
T-Stitch Pan et al., 2025	Which model runs each step	None	Preset switching point	NONE
Adaptive Guidance Castillo et al., 2025	Whether to run the unguided pass	Agreement of guided and conditional updates	Tuned threshold	NONE
Adaptive ASR correction Damri & Laufer-Goldshtein, 2026	Hypotheses per input	ASR confidence scores	Calibrated threshold	RISK CONTROL (LTT)
This study	Teacher-to-student switching time	Prompt SNR + endpoint stabilization	Calibrated threshold	RISK CONTROL (LTT)

SOURCES Graves (2016) · Teerapittayanon et al. (2016), ICPR · Schuster et al. (2022), NeurIPS · Starodubcev et al. (2024), CVPR · Pan et al. (2025) · Castillo et al. (2025) · Damri & Laufer-Goldshtein (2026) · Proposal Table 2.1

SECTIONS 2.4–2.8 · LITERATURE SYNTHESIS

WHERE PRIOR WORK LEAVES THE GAPS


PER-INPUT ADAPTIVE
CALIBRATED RISK BOUND
1 ACT
BranchyNet
Adaptive Guidance
2 CALM
ASR correction
Learn-Then-Test
the tool, not a handover
THIS STUDY
1 Adaptive collaboration
none found
3 DMOSpeech 2
T-Stitch
TEACHER → STUDENT HANDOVER

1 Adaptive, but no bound on quality loss

Per-input rules built on a learned halting unit or a hand-tuned threshold. None states how much quality the saving may cost.

2 Bounded, but not for a handover

CALM and adaptive ASR correction calibrate a threshold with risk control, for exit layers and recognition hypotheses, not for a teacher-to-student switch in speech synthesis.

3 A handover, but one fixed for all inputs

DMOSpeech 2 switches at one time for every utterance; T-Stitch uses a preset switching point. Neither adapts to the input or bounds the risk.

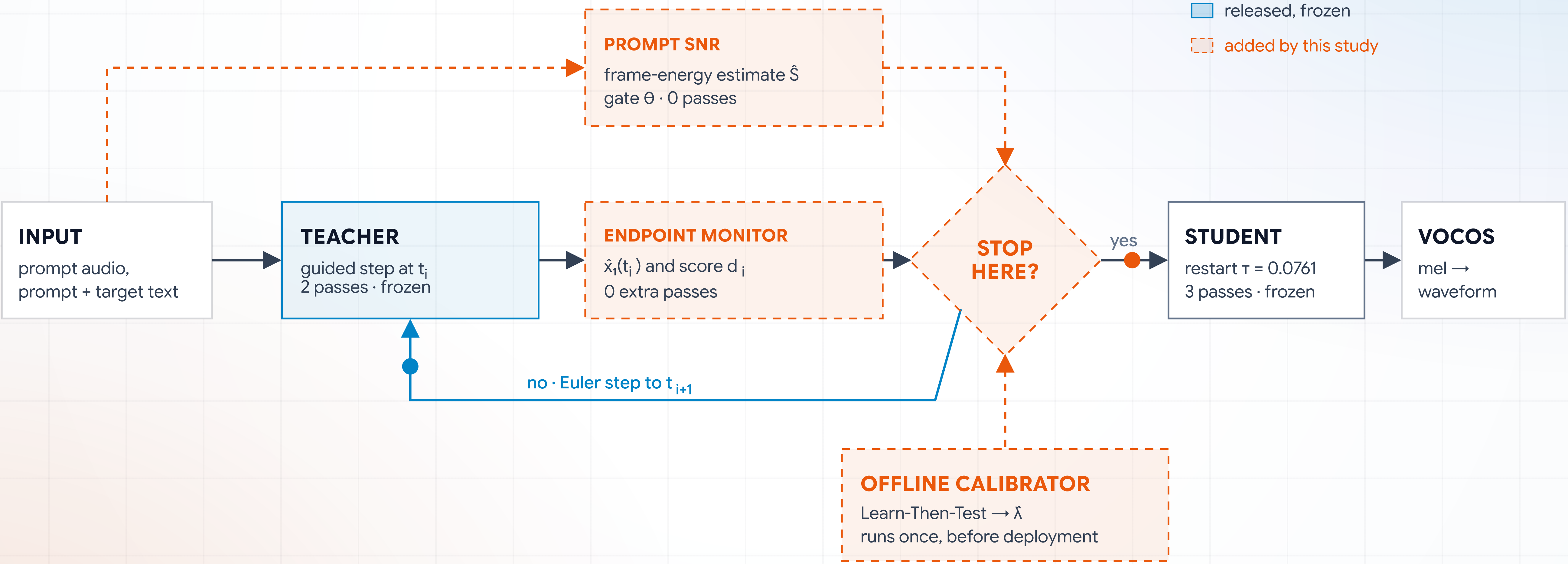
★ The centre is unoccupied

No reviewed method pairs a per-utterance handover with a calibrated bound. This describes the sources examined, not every adaptive method that exists.

SOURCES Graves (2016) · Teerapittayanon et al. (2016) · Schuster et al. (2022) · Starodubcev et al. (2024) · Pan et al. (2025) · Castillo et al. (2025) · Damri et al. (2026) · Li et al. (2026) · Angelopoulos et al. (2025) · Proposal Chapter II

SECTION 3.2 · SYSTEM ARCHITECTURE

MODULAR ADAPTIVE HANDOVER ARCHITECTURE



▭ released, frozen

⬚ added by this study

PROMPT SNR
frame-energy estimate $\hat{S}$
gate θ · 0 passes

INPUT
prompt audio,
prompt + target text

TEACHER
guided step at t_i
2 passes · frozen

ENDPOINT MONITOR
$\hat{x}_1(t_i)$ and score d_i
0 extra passes

STOP HERE?

yes

STUDENT
restart $\tau = 0.0761$
3 passes · frozen

VOCOS
mel ⟶
waveform

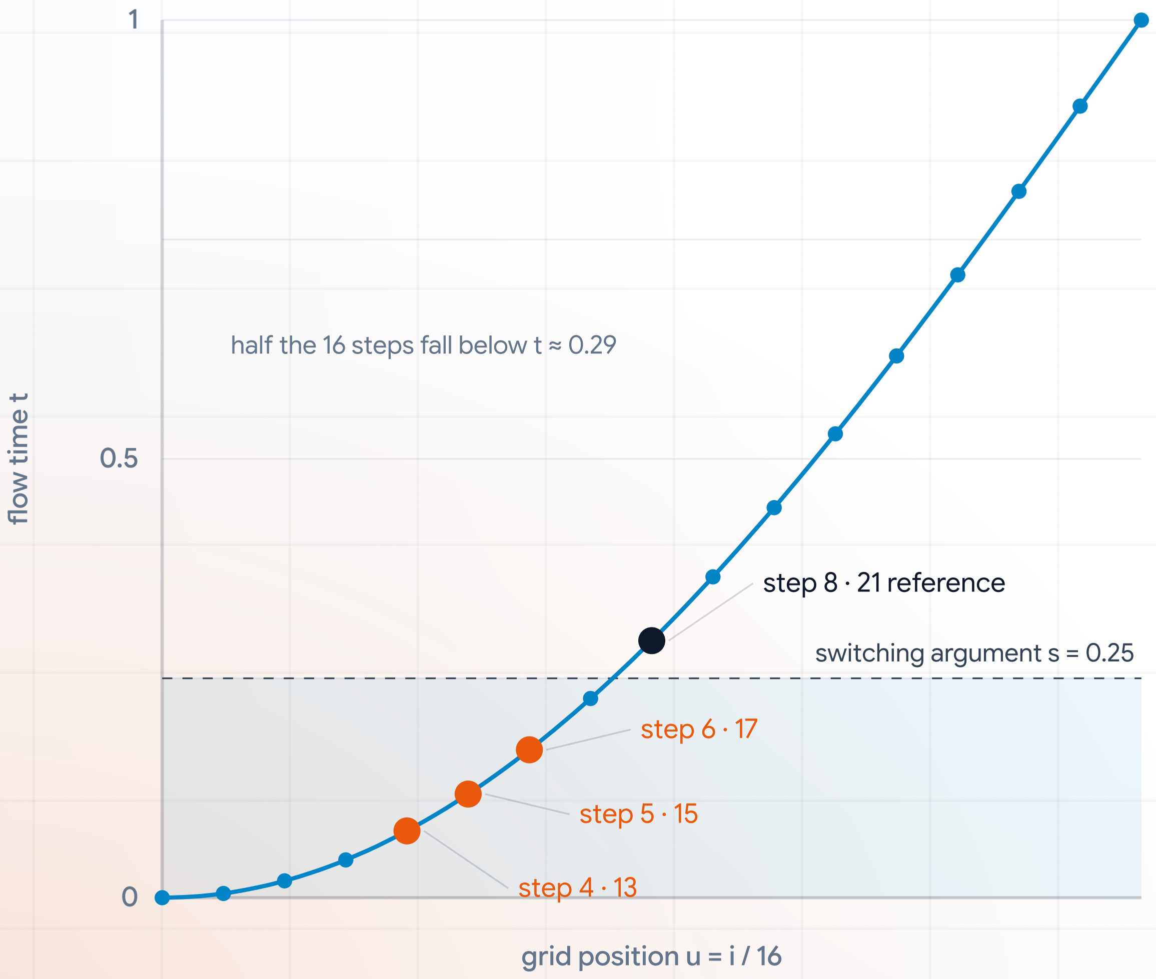
no · Euler step to t_{i+1}

OFFLINE CALIBRATOR
Learn-Then-Test ⟶ $\hat{\lambda}$
runs once, before deployment

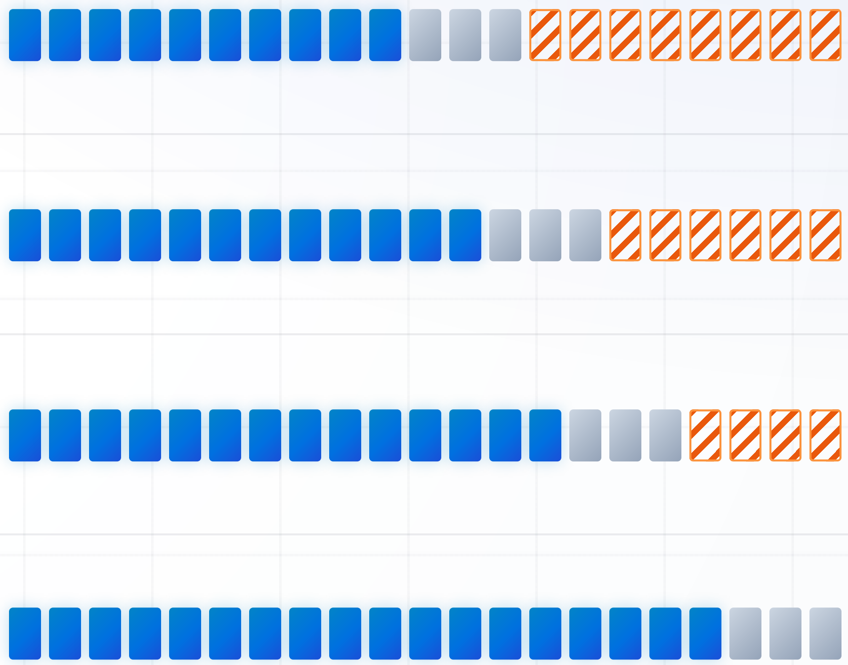
SOURCES Li et al. (2026), *DMOSpeech 2*, AAAI · Y. Chen et al. (2025), *F5-TTS* · Siuzdak (2024), *Vocos*, ICLR · Proposal Figure 3.2

SECTION 3.5 · CANDIDATE SWITCHING TIMES

SWAY SAMPLING DISCRETIZATION & CANDIDATE HANDOVER BOUNDARIES



Sway sampling packs teacher steps near t = 0. The released code stops at the **first grid point above** s. Cost: $C(s) = 2(k+1) + 3$.



s = 0.07	step 4 · t = 0.076	**13**	−8 · 38%
s = 0.10	step 5 · t = 0.118	**15**	−6 · 29%
s = 0.15	step 6 · t = 0.169	**17**	−4 · 19%
s = 0.25	step 8 · reference	**21**	baseline

$$x_\tau = (1-\tau)\,\varepsilon + \tau\,\tilde{x}, \quad \tau = 0.0761$$

student always re-noises the teacher’s estimate to the same level

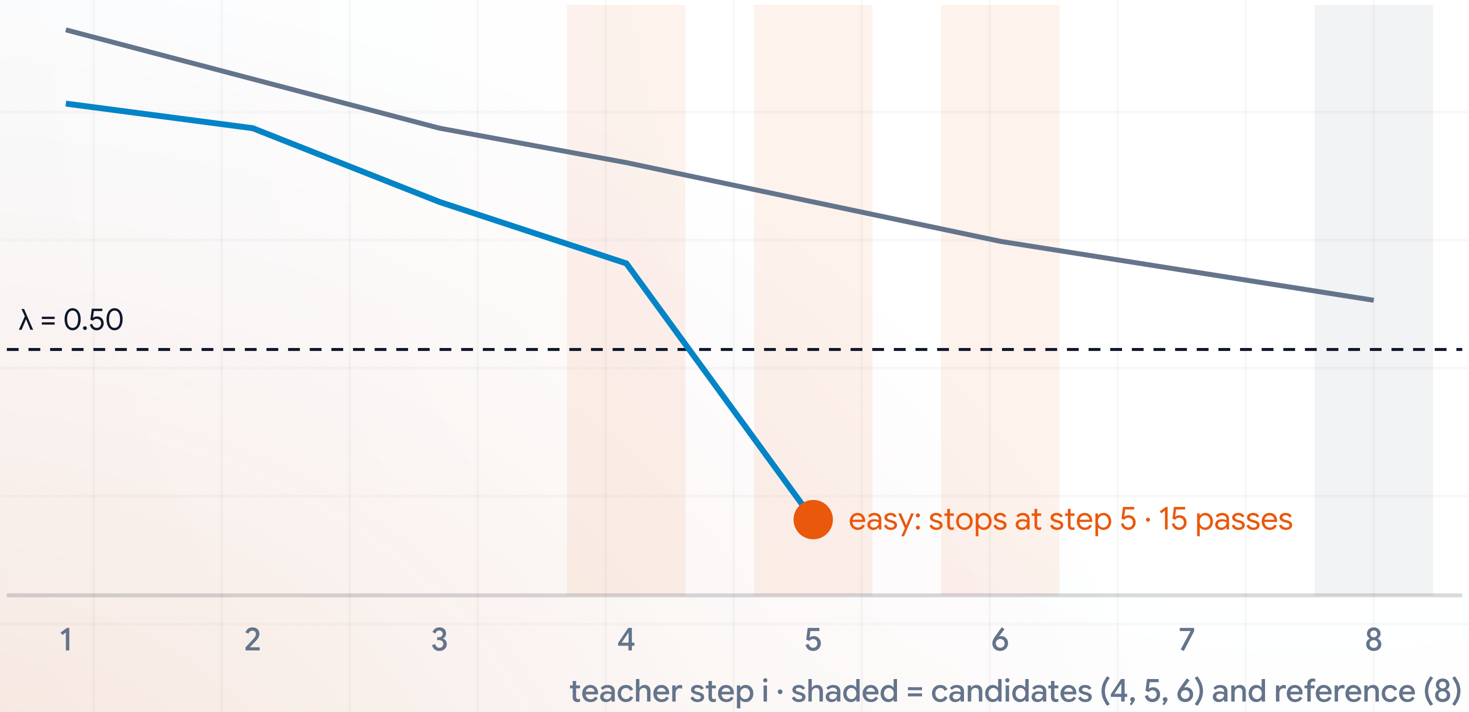
SOURCES Y. Chen et al. (2025), *F5-TTS*, ACL · Li et al. (2026), *DMOSpeech 2*, AAAI · Meng et al. (2022), *SDEdit*, ICLR · Xu et al. (2023), NeurIPS

SECTION 3.6.2 · DECISION FEATURES

ENDPOINT STABILIZATION & PROMPT ACOUSTIC QUALITY SIGNALS

$$d_i = \| \hat{x}_1(t_i) - \hat{x}_1(t_{i-1}) \|_{\mathrm{M}} \; / \; (\| \hat{x}_1(t_i) \|_{\mathrm{M}} + \epsilon), \quad \hat{x}_1 = x_t + (1 - t)\, v$$

endpoint-stabilization score · small means the teacher's guess has settled



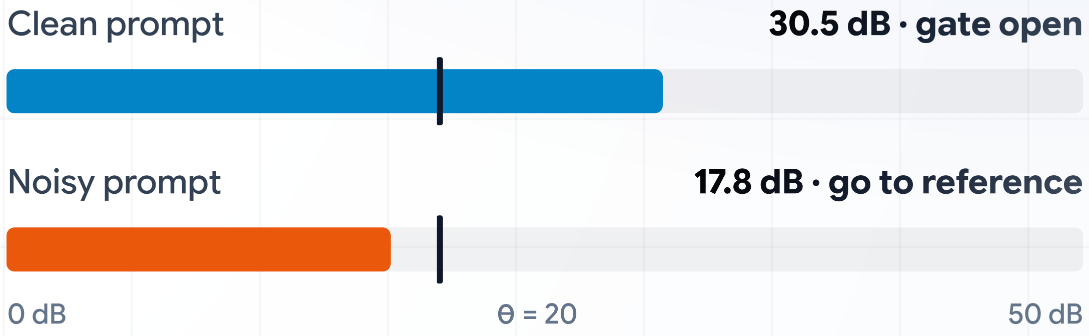
hard input · never settles → reference, 21 passes

easy: stops at step 5 · 15 passes

teacher step i · shaded = candidates (4, 5, 6) and reference (8)

$$\hat{S} = 10 \log_{10}(\bar{E}_{\mathrm{top}} \; / \; \bar{E}_{\mathrm{bottom}})$$

prompt SNR, from frame energies · gate θ = 20 dB



Clean prompt **30.5 dB · gate open**

Noisy prompt **17.8 dB · go to reference**

Development diagnostics suggest the **SNR may carry most of the benefit**. The stabilization score is kept as a hypothesis to test, not an assumption.

SOURCES Lipman et al. (2024), *Flow matching guide and code* · Li et al. (2026) · Proposal Def. 3.2–3.3, Appendix A.4 and D.4 (illustrative values)

SECTION 3.6 · MATHEMATICAL FORMULATION

FORMULATION OF THE CONSTRAINED STOPPING POLICY

$$\min_{\pi} \mathrm{E}[\ C(\pi(x))\ ] \quad \text{s.t.} \quad R(\pi) = \mathrm{E}[\ f(x, \pi(x))\ ] \le \alpha$$

cheapest policy whose added-failure risk stays under α

$$\pi_{\lambda}(x) = \text{first candidate } i_r \text{ with } d_{i_r} \le \lambda \text{ and } \hat{S} \ge \vartheta, \text{ else reference}$$

walks the teacher steps in order · never later than step 8

ADDED FAILURE · FIXED BEFORE CALIBRATION

$f = 1$ if WER > max(**12.5%**, $\mathrm{WER}_{\mathrm{ref}}$ + **1 pt**) or SIM < $\mathrm{SIM}_{\mathrm{ref}}$ − **0.02**, on the same seeds.

Setting $\lambda = \infty$ gives the SNR-only rule; $\theta = -\infty$ gives stabilization only. **One family, one calibration.**

PROP. 3.2

ZERO OVERHEAD

Signals reuse the velocity already computed. No extra forward pass.

PROP. 3.3

MONOTONE COST

A smaller λ never stops earlier, so cost rises as λ falls.

PROP. 3.4

SAFE FALLBACK

Never more than 21 passes. No certified λ → exact reference output.

PROP. 3.5

ORACLE BOUND

No zero-risk policy beats the cheapest non-failing candidate per input.

FROM Proposal Eq. 3-11, 3-12, 3-15 and Propositions 3.2–3.5 · WER: Radford et al. (2023) · SIM: S. Chen et al. (2022)

SECTION 3.7 · RISK CALIBRATION PROTOCOL

RISK-CALIBRATED THRESHOLD SELECTION VIA LEARN-THEN-TEST

SPEAKER-DISJOINT SPLITS · SEED-TTS TEST-EN · ONE UTTERANCE PER SPEAKER

Development	Calibration	Test
100	200	200

ILLUSTRATIVE · $n = 100$, $\alpha = 0.05$, $\delta = 0.10$ · $p = P(\mathrm{Bin}(n, \alpha) \le \text{failures})$

	g = 1 · conservative	g = 2	g = 3	g = 4 · aggressive
	λ 0.40	**λ 0.50**	**λ 0.60**	**λ 0.70**
failures	0	1	3	6
p	0.0059	0.0371	0.2578	—
	≤ Δ · CERTIFY	≤ Δ · CERTIFY	> Δ · STOP	NOT TESTED

$\hat{\lambda} = 0.50$

Guarantee: **$P(R(\hat{\lambda}) \le \alpha) \ge 1 - \delta$**, on average over new speakers. With zero failures, certifying $\alpha = 0.05$ at $\delta = 0.10$ needs **45** inputs (Hoeffding: ≈461). Protocol fixed in advance: grid 0.20–1.20 in steps of 0.05; verified on the **held-out test split** and **100 re-splits**.

SOURCES Angelopoulos et al. (2025), *Learn then test* · Bauer (1991), Stat. Med. · Clopper & Pearson (1934) · Hoeffding (1963) · Proposal Alg. 3.4, Appendix A.6

SECTIONS 3.1, 3.3–3.8 · RESEARCH DESIGN & PROTOCOL

EXPERIMENTAL WORKFLOW, BASELINES, & EVALUATION METRICS

SETUP
ALL SPLITS

- 1 Speaker splits
- 2 Clean + 15 dB noisy
- 3 Check seed, fallback, pass count

OBJECTIVE 1
DEVELOPMENT · EXPLORATORY

- 4 One teacher run, all candidates
- 5 Score + failures
- 6 Quality–cost map, fix θ

OBJECTIVE 2
NO DATA

- 7 Problem, policy, proofs

OBJECTIVE 3
CALIBRATION, TEST · CONFIRMATORY

- 8 Fix α, δ, grid
- 9 Binomial tests in order
- 10 Lock $\hat{\lambda}$ or fall back
- 11 Held-out + 100 re-splits

OBJECTIVE 4
TEST · CONFIRMATORY

- 12 Run locked variants
- 13 Paired comparisons
- 14 Noise shift, n = 50/100/200
- 15 Report all, incl. negative

COMPARED METHODS

Fixed $s \in \{0.07, 0.10, 0.15, \mathbf{0.25}\}$ · SNR-only · SNR + fixed later step · stabilization only · **full policy** · cost-matched random · oracle

MEASURES

WER Whisper-large-v3 · **SIM** WavLM-Base+ SV · **UTMOS** · exact **forward passes** · runtime and RTF on the T4

STATISTICS

Wilcoxon signed-rank + Holm · exact McNemar · TOST for equivalence · Clopper–Pearson intervals · bootstrap CIs

SOURCES Radford et al. (2023) · S. Chen et al. (2022) · Saeki et al. (2022), *UTMOS* · Wilcoxon (1945) · Schuirmann (1987)

CONTRIBUTION & SUMMARY

WHAT THIS STUDY ADDS

Within the sources reviewed, no method combines all three for flow-matching TTS.

	PRIOR WORK	THIS STUDY
01 **When to hand over**	One switching time for every utterance: fixed at 0.25 in DMOSpeech 2, preset in T-Stitch.	Chosen **per utterance**, from signals the sampler already computes: the prompt's SNR and the endpoint-stabilization score.
02 **How sure we can be**	Adaptive rules use learned or hand-tuned thresholds with **no bound** on quality loss. Calibrated bounds exist, but for exit layers and ASR hypotheses.	A **Learn-Then-Test** threshold: added failures $\le \alpha = 0.05$ with probability $\ge 90\%$, checked on held-out speakers.
03 **What it costs**	Some need extra training, or a quality estimator run on a full output.	**No training and no extra forward pass.** With no certified threshold it falls back to the released sampler, so it never costs more.

NOT CLAIMED A proven advantage of the hybrid over its parts, any change to duration prediction, or any new model training.

FROM Proposal, Section 2.8, Section 3.7 and Propositions 3.2 and 3.4

APPENDIX D.1–D.3 · PRELIMINARY EXPERIMENTS

WHAT THE EARLIER RUNS SHOWED

D.1 · FIXED SWITCHING TIMES · 500 SPEAKERS × CLEAN AND 15 DB NOISY · TESLA T4

SWITCH	PASSES	CLEAN (500)	NOISY (500)	ALL (1,000)
s = 0.07	13	12 (2.4%)	**322 (64.4%)**	334 (33.4%)
s = 0.10	15	9 (1.8%)	**269 (53.8%)**	278 (27.8%)
s = 0.15	17	5 (1.0%)	**247 (49.4%)**	252 (25.2%)
s = 0.25 · ref.	21	0	0	0

A per-input choice in hindsight needs **14.85 of 21 passes (−29%)**; the 179 instances that needed the reference were all noisy.

D.2 · STUDENT RESTART LEVEL

Restarting at j = 2 departs from the j = 1 reference more often (26 vs 11 of 48 at s = 0.07; 14 vs 0 at s = 0.25, zero by construction). Small CPU run, not a test of absolute quality. **j = 1 is kept as a controlled setting.**

D.3 · STABILIZATION POLICY, CROSS-VALIDATED

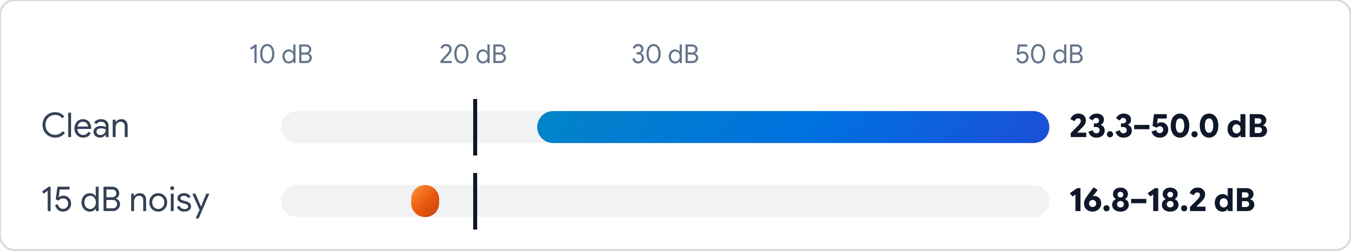
100 development speakers, four folds: **18.88 passes (−10.1%)** with 1 added failure in 200, against 50 for a fixed s = 0.15. Chosen with zero training failures, so **no risk certificate**; Learn-Then-Test replaces this selection.

FROM Proposal Appendix D.1–D.3 and Table D.1 · development evidence, not independent held-out validation

APPENDIX D.4–D.5 · DEVELOPMENT-SPLIT SIGNAL DIAGNOSTIC

HOW THE TWO SIGNALS BEHAVED

PROMPT SNR · 100 DEVELOPMENT SPEAKERS



The SNR **separated the two conditions completely**; a 20 dB gate sent every noisy prompt to the reference. Fixed early handovers failed on 49–63% of noisy prompts but 2–4% of clean ones.

STABILIZATION SCORE D_4 ON CLEAN PROMPTS

AUC ≈ **0.47** for Step-4 failures vs passes, from only two to three failures. At $\lambda = 0.50$ the policy waited past Step 4 on 78 of 100 clean prompts, of which 75–77 would already have passed.

D.5 · PLAN, FIXED BEFORE THE DATA ARE SEEN

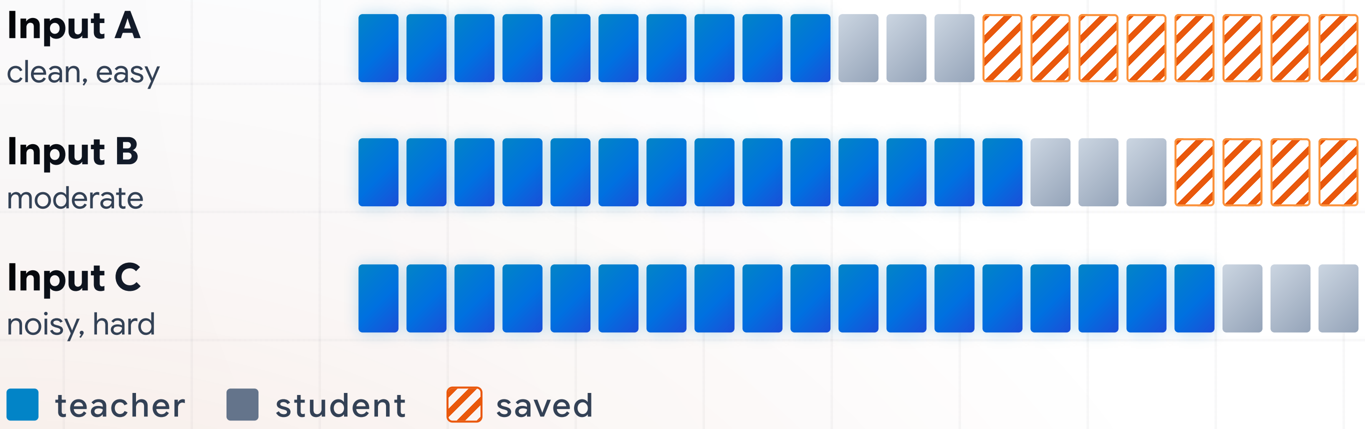
1 Discrimination on development · 2 Controls on test (SNR + fixed later step, cost-matched random) · 3 Prompts the SNR cannot see (reverberation, telephone band) · 4 Richer zero-cost signals. If the score adds nothing, the **SNR-gated policy** is recommended.

FROM Proposal Appendix D.4–D.5 · development split only; not evidence for or against the stabilization score

APPENDICES B–C · SIMULATION AND PROTOTYPE

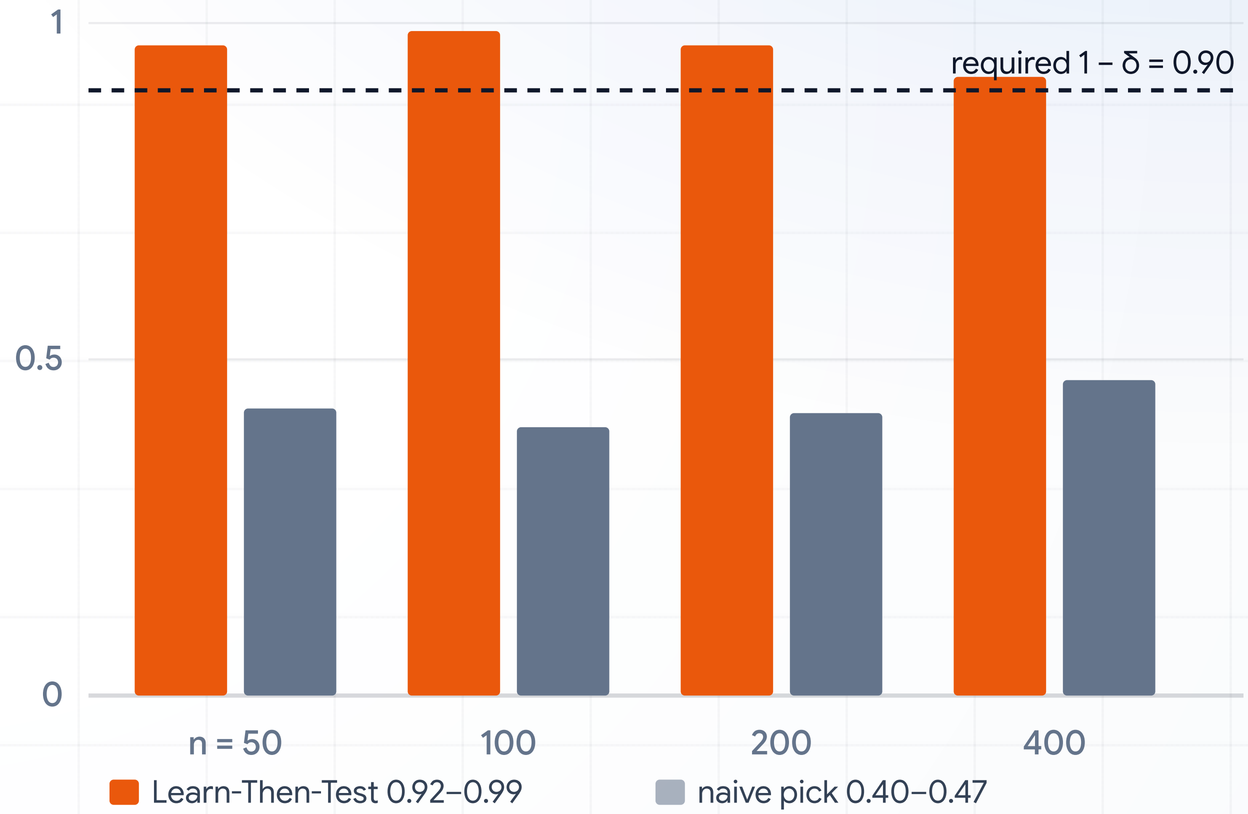
PROJECTED COMPUTATIONAL SAVINGS & SIMULATION FINDINGS

PROTOTYPE · HANDOVER PER INPUT


Input A
clean, easy
Input B
moderate
Input C
noisy, hard
🟦 teacher
■ student
▨ saved

Easy inputs save up to **38%**. Hard inputs cost **exactly the baseline**, never more.

SIMULATION · P(R ≤ α) OVER 1,000 RUNS



Simulated data, not experimental results. LTT used 15.9–18.2 passes on average vs 21 for the reference.

FROM Proposal Appendix B (Table B.1) and Appendix C (Figure C.1) · costs exact by Proposition 3.1